

# Valdoria Votes: A Political Mystery (Part 2)

## Question Menu

1. Investigate Alerts
2. Analyze Logs
3. Write Reports

<https://kc7cyber.com/challenges/335>

Your mission: investigate the events leading up to this critical election. Analyze the data, uncover any hidden agendas, and help protect the integrity of Valdoria's vote.

Tool: KQL (Kusto Query Language)

### What is the name of the Deputy Commissioner?

```
Employees  
| where role == "Deputy Commissioner"
```

Name: Hilary Binton

### What is Dora Thomas' role?

```
Employees  
| where name == "Dora Thomas"
```

Polling Station Supervisor

### What is this supervisor's name? Temp Election Support Staff Supervisor

```
Employees  
| where role contains "supervisor"
```

Barry Schmelly

### What is Barry Schmelly's IP address?

```
Employees  
| where name == "Barry Schmelly"
```

10.10.0.12

### What is Barry Schmelly's hostname?

Hostname= GCH3-DESKTOP

### Email address

barry\_schmelly@valdoriavotes.gov

**How many emails did Barry Schmelly receive?**

```
Email
| where recipient == "barry_schmelly@valdoriavotes.gov"
| count
37
```

**How many distinct commands were run on Barry Schmelly's machine?**

```
ProcessEvents
| where hostname == "GCH3-DESKTOP"
| distinct process_commandline
```

173

**How many distinct URLs did employees with the first name William visit?**

```
let wills_ips =
Employees
| where name has "William"
| distinct ip_addr;
OutboundNetworkEvents
| where src_ip in (wills_ips)
| distinct url
```

217

**How many authentication attempts did we see to the accounts of employees with the first name William?**

```
let w_username = Employees
| where name has "William"
| distinct username;
AuthenticationEvents
| where username in (w_username)
```

183

When in doubt you can do a `take 10` to quickly peek into

```
let w_username = Employees
| where name has "William"
| distinct username;
AuthenticationEvents
| where username in (w_username)
| take 10
```

## Which table would tell us if Barry Schmelly received a suspicious email?

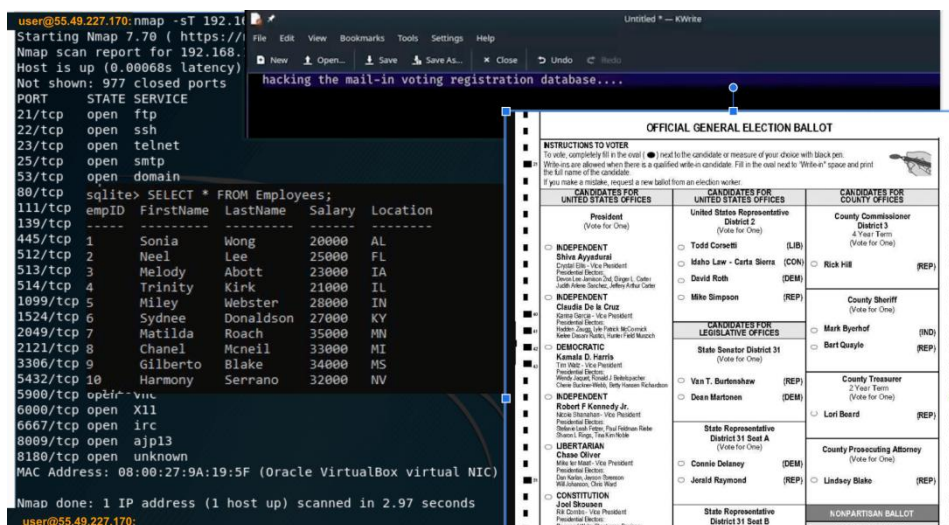
| Table Name                                                                                              | Description                                           | Example                                                   |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|
|  AuthenticationEvents  | Logins to devices (successes and failures).           | When someone tries to log in to their email account.      |
|  Email                 | Emails sent and received by employees.                | When an employee sends or receives a work email.          |
|  Employees             | Info about the company's employees.                   | Names, job titles, and contact details of employees.      |
|  FileCreationEvents    | Files created on employee devices.                    | When someone saves a new document on their computer.      |
|  InboundNetworkEvents  | Internet activity coming into the company network.    | When someone visits the company website.                  |
|  OutboundNetworkEvents | Internet activity going out from the company network. | When someone visits an external website like google.com.  |
|  PassiveDns (External) | Which IP addresses match up with which websites.      | Figuring out that "google.com" matches with IP "8.8.8.8". |
|  ProcessEvents         | Programs that start running on employee devices.      | When an employee opens a program like Microsoft Word.     |
|  SecurityAlerts        | Security warnings from devices or email systems.      | When antivirus software detects a virus on a computer.    |

Table very useful to be translate to other sequel language

An unknown hacking group claims to have compromised the voting machines in Valdoria. But how? The voting machines are supposed to be extremely hard to hack! The voting systems are airgapped. So how did they actually do it? How could they have done it?

## What IP address did they forget to hide in the poster?

55.49.227.170



The screenshot shows a Kali Linux terminal window with the following output:

```

user@55.49.227.170: nmap -sT 192.168.1.100
Starting Nmap 7.70 ( https://nmap.org )
Nmap scan report for 192.168.1.100
Host is up (0.00068s latency)
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  smb
445/tcp   open  smb
512/tcp   open  ftp
513/tcp   open  ftp
514/tcp   open  ftp
1099/tcp  open  http
1524/tcp  open  http
2049/tcp  open  http
2121/tcp  open  http
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  http
6000/tcp  open  http
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:9A:19:5F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.97 seconds
user@55.49.227.170:

```

The terminal also shows a sqlite query and its results:

```

sqlite> SELECT * FROM Employees;
empID  FirstName  LastName  Salary  Location
-----
1       Sonia      Wong      20000   AL
2       Neel       Lee       25000   FL
3       Melody     Abott     23000   IA
4       Trinity    Kirk      21000   IL
5       Miley      Webster   28000   IN
6       Sydnee     Donaldson 27000   KY
7       Matilda    Roach     35000   MN
8       Chanel     Mcneil    33000   MI
9       Gilberto   Blake     34000   MS
10      Harmony    Serrano   32000   NV

```

The ballot paper is for the 'OFFICIAL GENERAL ELECTION BALLOT' and lists candidates for various offices including President, United States Representative, County Commissioner, etc.

Is there any evidence of traffic from this IP address to our network? (yes/no)

No

We need a ping, only shown are open port with the tool Nmap

**How many domains resolved to the IP 55.49.\*\*.\*\*\*?**

```
PassiveDns
| where ip contains "55.49."
```

55.49.227.170

Domains found

valdoriavotesgov.com

shadow-hackers-r.us

**What is the Valdoria related domain resolving to that IP?**

valdoriavotesgov.com

**What is the legitimate domain for the Valdoria Elections Board?**

valdoriavotes.gov

**How many IP addresses did the fraudulent Valdoria government domain resolve to?**

```
PassiveDns
| where domain contains "valdoriavotesgov.com"
```

214.85.104.248

55.49.227.170

157.100.244.104

**Which table should we be looking at for records of that kind of activity again?**

InboundNetworkEvents any browsing activity from these IPs to our network

**How many requests do we see to our network from those IPs?**

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
```

26

**Which phase of the Cyber Kill Chain is this called?**

recon;reconnaissance

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
```

```
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
| where url contains "prevention"
election interference
```

**What important machines were the hackers trying to get into? (two words)**

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
| where url contains "machine"
voting machines
```

**Which document were the hackers looking for? (two words)**

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
| where url contains "document"
```

**Which document were the hackers looking for? (two words)**

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
```

technical manual

**Let's check if there's any traffic to it—has any of our employees visited that domain for any reason? (yes/no)**

```
OutboundNetworkEvents
| where url contains "valdoriavotesgov.com"
```

Yes 10.10.0.4

```
Employees
| where ip_addr == "10.10.0.4"
```

Anderson Snooper

**When did someone first browse to that domain?(paste the full timestamp)**

2024-10-07T10:46:45Z

Did anyone enter their credentials on that phishing page ? (yes/no)

Yes

2024-10-07T10:46:47.000Z

What is the username of the employee that entered their credentials on that phishing page?

Employees

| where ip\_addr == "10.10.0.4"

ansnooper

Anderson Snooper

Temp Election Support Staff Lead

False

What table would we find this logon activity in?

AuthenticationEvents

| Table Name                                                                                                | Description                                           | Example                                                   |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|
|  AuthenticationEvents  | Logins to devices (successes and failures).           | When someone tries to log in to their email account.      |
|  Email                 | Emails sent and received by employees.                | When an employee sends or receives a work email.          |
|  Employees             | Info about the company's employees.                   | Names, job titles, and contact details of employees.      |
|  FileCreationEvents    | Files created on employee devices.                    | When someone saves a new document on their computer.      |
|  InboundNetworkEvents  | Internet activity coming into the company network.    | When someone visits the company website.                  |
|  OutboundNetworkEvents | Internet activity going out from the company network. | When someone visits an external website like google.com.  |
|  PassiveDns (External) | Which IP addresses match up with which websites.      | Figuring out that "google.com" matches with IP "8.8.8.8". |
|  ProcessEvents         | Programs that start running on employee devices.      | When an employee opens a program like Microsoft Word.     |
|  SecurityAlerts        | Security warnings from devices or email systems.      | When antivirus software detects a virus on a computer.    |

When did the threat actor login to Snooper's account?

From the previous question we know that valdoriavotesgov.com as 3 SRC\_IP

214.85.104.248

55.49.227.170  
157.100.244.104

So we can write the SQL

```
AuthenticationEvents
| where username == "ansnooper"
| where timestamp between(datetime(2024-10-05T10:46:47Z) .. datetime(2024-10-11T10:46:47Z))
| where src_ip contains "214.85.104.248" or src_ip contains "55.49.227.170" or src_ip contains "157.100.244.104"
```

We found 157.100.244.104 was reached

2024-10-07T15:46:45.000Z

### What is Anderson Snooper's email address?

```
Employees
| where ip_addr == "10.10.0.4"
```

anderson\_snooper@valdoriavotes.gov

On October 8th, 'Snooper' had an interesting email conversation that's worth investigating

### What is the email address of the person he was conversing with?

```
Email
| where recipient == "anderson_snooper@valdoriavotes.gov"
| where timestamp between ( datetime(2024-10-08)..datetime(2024-10-09))
| where subject contains "barry_schmelly@valdoriavotes.gov"
```

### What is Barry Schmelly's IP address?

```
Employees
| where name == "Barry Schmelly"
```

### What is Schmelly's job role?

Temp Election Support Staff Supervisor

### "Snooper" was observed asking Schmelly how one might gain access to what devices?

```
Email
| where recipient == "anderson_snooper@valdoriavotes.gov"
| where timestamp between ( datetime(2024-10-08)..datetime(2024-10-09))
| where subject contains "machine"
```

voting machines

Email

```
| where sender == "anderson_snooper@valdoriavotes.gov"
| where recipient == "barry_schmelly@valdoriavotes.gov"
| where timestamp between ( datetime(2024-10-08)..datetime(2024-10-09))
```

do my job though

AI system

## What is Snooper's IP address?

OutboundNetworkEvents

```
| where url contains "valdoriavotesgov.com"
10.10.0.4
```

Employees

```
| where ip_addr == "10.10.0.4"
```

ansnooper

Anderson Snooper

Temp Election Support Staff Lead

## What term appeared at the end of each url that Snooper guessed?

```
let Snooper_ips =Employees
| where name has "Snooper"
| distinct ip_addr;
InboundNetworkEvents
| where src_ip in (Snooper_ips)
| distinct url
```

model=gpt-4o

```
let Snooper_ips =Employees
| where name has "Snooper"
| distinct ip_addr;
InboundNetworkEvents
| where src_ip in (Snooper_ips)
```

Select early time

2024-10-16T15:57:05.000Z

Response= Ai

## What status code was returned for Snooper's unsuccessful guesses?

Response= 404

## What was the nursery rhyme-related subdomain that Snooper guessed?

ol-mcdonald-had-a-farm-ai-ai-oh

**What was the basketball player-related subdomain that Snooper guessed?**

allen-iverson

**What was the first subdomain "Snooper" guessed that returned a 200 status code?**

https://elections-chatbot.valdoriavotes.gov/?model=gpt-4o

elections-chatbot

**How many questions did they ask the chatbot?**

```
let Snooper_ips =Employees
| where name has "Snooper"
| distinct ip_addr;
InboundNetworkEvents
| where src_ip in (Snooper_ips)
| where url contains "elections-chatbot.valdoriavotes.gov/?model=gpt-4o?prompt"
```

Response=6

By checking the AIPrompts table, we can find out what response the AI chatbot gave.

**Which conversation\_id is associated with the question about voting machines?**

```
AIPrompts
| where prompt has "voting machines"
| project conversation_id
```

94bd6162-1323-402d-bccd-8fceaee5f230

Let's "pivot" on that conversation\_id to see what else that user searched for.

**The AI bot told the threat actor to bring a \_ and a banana.**

```
AIPrompts
| where conversation_id == "94bd6162-1323-402d-bccd-8fceaee5f230"
| where response contains "banana"
```

bring a flashlight and a banana

flashlight

According to the bot, the voting machines are not actually connected to the internet.

**Instead, votes are manually calculated using a \_\_\_\_.**

```
AIPrompts
| where conversation_id == "94bd6162-1323-402d-bccd-8fceaee5f230"
| where response contains "calcul"
```

Calculator

After trolling the threat actor for a bit, the AI chatbot suggests they might have better luck getting information about the voting machines from the vendor who makes them.

**What is the name of the vendor?**

```
AIPrompts
| where conversation_id == "94bd6162-1323-402d-bccd-8fceaee5f230"
| where response contains "vendor"
```

Voting vendor= Dominos Voting Systems

Election Commissioner

And who is that?

**What is the name of the employee with that job role?**

```
Employees
| where role == "Election Commissioner"
```

Name = Arrack Bobama

Username= arbobama

Following the password reset, the threat actors logged into Bobama's account.

**When did they log in to Bobama's account?**

```
let w_username = Employees
|where name has "Bobama"
|distinct username;
AuthenticationEvents
| where username in (w_username)
| where src_ip contains "214.85.104.248" or src_ip contains "55.49.227.170" or src_ip
contains "157.100.244.104"
| take 10
```

2024-10-16T00:00:00.000Z

**What IP address did the threat actors use to log in to that account?**

Src\_ip= 214.85.104.248

The threat actors (using Bobama's account) were later observed sending an email to the voting machine vendor.

### What email address did they send this email to?

```
Employees
| where role == "Election Commissioner"
```

Ip\_addr= 10.10.0.13

Email: arrack\_bobama@valdoriavotes.gov

Voting vendor= Dominos Voting Systems

```
Email
| where sender == "arrack_bobama@valdoriavotes.gov"
| where recipient contains "DominosVotingSystems"
| where timestamp > datetime(2024-10-16)
```

help@dominosvotingsystems.com

In this conversation, the threat actors are once again informed that the voting machines can only be accessed physically, as they are not connected to the internet.

### What PDF did the threat actors receive that might be useful to them later?

```
Email
| where recipient contains "arrack_bobama@valdoriavotes.gov"
| where attachments contains "pdf"
| where timestamp > datetime(2024-10-16)
```

ValdoriaVotingMachinesNetworkGuide.pdf

## What You'll Learn

deal with yet another election drama in Valdoria 🗳️

Core Analyst Competencies

Choosing the Right Data Source

Documenting Your Work

Recognizing Investigation Starting Points

Querying Logs

Pivoting Across Data

Catching Your Own Blind Spots

Forming and Testing Hypotheses

Spotting Abnormal Activity

Domain-Specific Knowledge

Mapping Attacker TTPs

Recognizing Common Entry Points

Understanding Malicious Email Content  
Understanding How SIEMs Work  
Understanding Active Directory  
Reading Process Trees  
Advanced Topics  
Handling Cloud Forensics Challenges

|                       |               |
|-----------------------|---------------|
| // DETAILED ANALYSIS  |               |
| TOTAL_SCORE           | 2750 pts      |
| CHALLENGES_SOLVED     | 67 / 67       |
| TOTAL_ATTEMPTS        | 114           |
| AVG_TIME_PER_QUESTION | 1h 13m 19.4s  |
| COMPLETION_TIME       | 3d 9h 52m 39s |